

THE ICE CAVES · THE FEROCIOUS YETI · EVERY UNIT, EVERY DIFFICULTY

What every cave yeti, spearman, fisher, dark yeti swordsman and THE FEROCIOUS YETI does to your men on EASY, MEDIUM, HARD and MEGA, with every body's HP and head-count · read out of the live game v29.00 on Nov 1, 2026 16:01 EST · every damage number checked on the game's own machines (Chrome 381/381 · WebKit 381/381) · for Raziya

The one thing to know: in the caves **three blows kill a foot soldier whatever his HP**: a swordsman's cut, THE FEROCIOUS YETI's landing (THE CHOMP, 58 px wide) and the falling ice (which kills **everyone** under it: Sarge of every iron, the jeep, the tank, every hero). Every other blow is **bar hits** off the one bar every man carries (7 for the men, 6 for the medic, 12 for the saber Sarge) × the share: a cave yeti's landing, a spearman's charge, a fisher's fish and the stomp 1 · 1 · 2 · 2 by tier, the chase's run 2, a cut on Sarge or the medic 3, THE CHOMP on them 2 · 3 · 4 · 4. **So the heavies are not heavy here**: a flame (51 HP) and a rifleman (22 HP) fall to the same cut, the same landing, the same shard, and stand the same 7 landings. **King Yeti and King Rex are the exception the other way**: every cave creature's blow bounces off their hide (measured: nothing lands but the ice). Difficulty changes the bar hits (1 → 2 from HARD), THE CHOMP (2 · 3 · 4 · 4), the bodies (×1 · 1.5 · 2.2 · 2.64), the head-counts, his bar (1,080 · 1,890 · 1,881 · 3,089) and the lab's level.

THE BOSS LAB fights the caves with its stock army at **LEVEL 45 on EASY, 47 on MEDIUM, 48 on HARD and 50 on MEGA**: Sarge with the saber, the jeep, the tank and the dogfighter (the bomber on MEGA), the jet pack on, and eight men at that rank (rifleman, M. gunner, bazooka, mortar, medic, flame, sweeper, radio man; MEGA swaps the radio man for the bomber). The grenadier, the commando and the spy are a tap away in THE PLATOON. Every "level" number on this page is that army at that tier's level. At levels 45 to 48 the men's HP is the same number (the ladder steps every tenth level); 50 is a step up.

What changed recently (since Sep 22)

live since	when	card	what it does in this fight
v29.00	Nov 1	**CARD-ICN-G3 · THE HOLES ARE REAL**	A hole's rim steers a man's step, a man pushed across a rim falls in (the pit law), and the hand is refused at a rim.

live since	when	card	what it does in this fight
v28.99	Nov 1	**CARD-ICN4 · THE CAVES' PURSE**	A cave creature pays its bounty × 4 · 3.6 · 3.2 · 3 dog tags by tier (a cave yeti 40 · 36 · 32 · 30, a swordsman 120 · 108 · 96 · 90; measured as born); a man THE CAVE-IN's band kills gets ×1 · 1 · 0.75 · 0.5 of his price back in all.
v28.98	Nov 1	**CARD-ICN-G1 · THE HOLES IN VIEW**	The caves pan down for the hand like the flat rugs, so the bottom row of holes clears the troop tray on a phone.
v28.97	Nov 1	**CARD-ICN-G2 · THE QUIET START**	The first burst at 8 · 6 · 5 · 4 s by tier (was 2.5 everywhere), the three holes nearest the toy box asleep till 25 · 20 · 15 · 10 s (never past his arrival), and the lab forgets the speed dial on a reload.
v28.74	Oct 31	**CARD-HP3 · THE HEAVIES**	+3 HP every tenth rank for the bazooka, grenadier, flame and commando. At level 50: bazooka 49, grenadier 45, flame 61, commando 58 HP; at 45 to 48: 40 · 37 · 51 · 49. A bar hit lands a hair lighter on them (the share); the three kill doors do not care.
v28.65	Oct 30	**CARD-LRB · ONE HP NUMBER**	Your man's HP is the number on his unit sheet; a bar hit lands 90 % + 10 % × bar ÷ HP of itself (a level-45 flame takes 0.91 of a cave yeti's 1, a rifleman 0.93).
v28.46	Oct 30	**CARD-BLP · THE BOSS LAB'S STOCK ARMY**	The lab fights the caves with its own army at LEVEL 45 · 47 · 48 · 50 (the lab's level sheet for the caves reads 45–50, spread over the four tiers): the one this page reads.
read	Oct 10	**CARD-BHP · ONE BOSS, MORE HEALTH**	A party grows his bar (×1 + 0.9 a commander past the first) instead of multiplying him.
read	Oct 8	**CARD-ICB · A FIGHT WITH NO DARK**	His bar re-priced for a fight all on the floor: ×2.4 · 2.8 · 1.9 · 2.6 by tier (was 0.8 · 2.24 · 1.52 · 2.08 before Oct 3's –20 % and this).
read	Oct 5	**CARD-CIR · THE COMMANDO'S ROOF**	A commando's shield breaks a falling shard over him or over any man within 120 px of him, once (his charge), measured on every tier.
read	Oct 3	**CARD-CVX · HIS BAR –20 %**	Wyatt 10/3: the Ferocious Yeti's health down 20 % on every difficulty; the fisher throws at a plane nearer than his man.
read	Oct 2	**CARD-HRT-C3 · THE PACE**	The cave-in's band walks its span in 9 · 13 · 16 · 18 s by tier; MEGA's swordsmen 1 a call, 2 live; MEGA's roar wakes 2 cave yetis.
read	Sep 26	**CARD-YF1 · YF3 · YF4 · CF · THE HITS LAND**	The swordsman's cut downs a foot soldier; THE CHOMP (his landing, 58 px wide) kills the foot soldiers under it and the rampage is his default landing (3 · 4 · 5 · 6 bounds); the falling ice kills (Wyatt 9/25: "the stalagmites, when they fall, need to destroy just about any unit"); the storm's pace by tier.
read	Sep 24	**CARD-YB · YR · THE SPEARMAN, THE FISHER, THE STORM**	The spearman's charge and the fisher's fish join the cave yetis; the stalactite storm aimed at the men.

live since	when	card	what it does in this fight
read	Sep 22	**CARD-DY · IBC · THE SWORDSMAN, THE RAIDERS**	The dark yeti swordsman (the lunge, three cuts) and the raiders who take the toy box.

How a hit is counted

Your men carry **HP**: the number on their unit sheet (a level-45 flame has 51, a rifleman 22; at level 50: 61 and 26). Their health bar is also cut into a few **bar hits** (7 for most men at these levels, 6 for the medic and the spy, 8 for the commando, 12 for the saber Sarge, 23 for the Claymore Sarge). Two kinds of blow in this fight:

- **Kill blows** (a swordsman's cut and THE CHOMP on a foot soldier: rifleman, M. gunner, bazooka, mortar, grenadier, sweeper, flame, radio man, commando, spy; the falling ice on anyone): out in one, whatever his HP. Only a wall turns them: a pet's shield, Francis's block, the commando's roof, the blue wall or the dome turns the blow into bar hits (1 · 1 · 2 · 2 for the cut, 3 for a shard); the champion's grace once a life.
- **Bar hits** (a landing, a charge, a fish, the stomp: 1 · 1 · 2 · 2; the run: 2; a cut on Sarge, the medic, a machine or a hero: 3; THE CHOMP on them: 2 · 3 · 4 · 4): since v28.65 a bar hit lands 90 % + 10 % × bar ÷ HP, so a level-45 rifleman takes 0.93 of his 7 from a cave yeti's landing (2.9 HP of 22). A hero's share is 1. A man who takes no hit for 4 s mends one bar hit, and again every 4 s.

Saves still roll on every bar hit: a jetting, dropping or dragged man cannot be touched, the pet shield, the deflect dome, the absorbs, and **the saber Sarge's block turns a landing, a charge, a fish, a cut or THE CHOMP about half the time** (measured: 4–7 of 10 landings, 2–8 of 10 cuts and 2–5 of 10 chomps turned over the four tiers and both engines). **King Yeti's and King Rex's hide turns every blow of a cave creature** (a thrown-snow law: the whole cave family counts as snow), the falling ice alone takes them. The Claymore Sarge's counted block and the riot plate keep their say. Nothing turns the ice but a shield.

The strikes · what a hit does by difficulty

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
A CAVE YETI'S LANDING · bar hits	**1 × share**	**1 × share**	**2 × share**	**2 × share**	every man within 34 px of where he lands	unchanged since Sep 24
A SPEARMAN'S CHARGE · bar hits	**1 × share**	**1 × share**	**2 × share**	**2 × share**	every man within 38 px of his line, once a charge	unchanged since Sep 24
A FISHER'S FISH · bar hits	**1 × share**	**1 × share**	**2 × share**	**2 × share**	every man within 30 px of the landing	unchanged since Sep 24
A SWORDSMAN'S CUT · a foot soldier	**KO**	**KO**	**KO**	**KO**	the nearest uncut man within 62 px in front of each cut, three cuts a lunge	since Sep 26 (the hits land)
A SWORDSMAN'S CUT · Sarge, the medic, the machines, a hero	**3 × share**	**3 × share**	**3 × share**	**3 × share**	a bar hit through his saves (the saber turns half; the hide of King Yeti and King Rex turns all)	since Sep 26
THE CHOMP · a foot soldier	**KO**	**KO**	**KO**	**KO**	every foot soldier within 58 px of his landing	since Sep 26 (a landing that kills)
THE CHOMP · Sarge, the medic, the machines, a hero	**2 × share**	**3 × share**	**4 × share**	**4 × share**	the bar hits and the bowl-over (80 px); a rampage's follow-up landing one bar hit lighter (1 · 2 · 3 · 3)	unchanged since Sep 26
THE STOMP · bar hits	**1 × share**	**1 × share**	**2 × share**	**2 × share**	every man within 150 px of him at the end of a 0.35 s rise, and a bowl-over of 130 px	unchanged

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
THE CHASE'S RUN · bar hits	**2 × share**	**2 × share**	**2 × share**	**2 × share**	every man within 40 px of him, every 0.5 s, and a bowl-over	unchanged
THE FALLING ICE	**KO**	**KO**	**KO**	**KO**	every unit within 40 px of a shard's shadow when it lands (1.8 s after it appears), Sarge and the machines too; a shield turns it into 3 bar hits; a commando's shield breaks the first shard over him or his circle (120 px)	since Sep 26 (the storm kills) · the roof Oct 5
BODY HP · the cave creatures	**×1**	**×1.5**	**×2.2**	**×2.64**	the cave yeti, the spearman, the fisher and the swordsman: the raid curve alone	unchanged
HEAD-COUNTS	**2 a burst · 1 · 1 · 1 live**	**2 a burst · 2 · 1 · 2 live**	**3 a burst · 3 · 2 · 3 live**	**3 a burst · 4 · 2 · 2 live**	cave yetis an opening burst · spearmen · fishers · swordsmen live at a time (a party adds one)	Oct 2 (the pace)
DOG TAGS A CAVE CREATURE PAYS	**×4**	**×3.6**	**×3.2**	**×3**	a cave yeti 40 · 36 · 32 · 30 · a spearman 64 · 58 · 51 · 48 · a fisher 56 · 50 · 45 · 42 · a swordsman 120 · 108 · 96 · 90 · the boss 300 · 300 · 300 · 300 (measured as born)	Nov 1 (the caves' purse)

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
THE CAVE-IN'S REFUND	**×1**	**×1**	**×0.75**	**×0.5**	the share of his price a man the band kills gets back in all (every fallen man gets half back)	Nov 1
THE FEROCIOUS YETI'S BAR (one commander)	**1,080**	**1,890**	**1,881**	**3,089**	450 × 2.4 · 2.8 · 1.9 · 2.6 × the raid curve ×1 · 1.5 · 2.2 · 2.64; a party grows it ×(1 + 0.9 a commander past the first)	Oct 8 (a fight with no dark) · Oct 3 (–20 %)

share = 90 % + 10 % × the man's bar hits ÷ his HP (CARD-LRB). The caves' dials sit in THE BOSS LAB's RULES under THE ICE CAVES (28 cells: his bar, his strike clock, his chomp, the chase's pace, the dark, the hold, the eyes, the calls, the bursts, a landing, the storm, the bounce chain, the swarms, the cave-in, the march, the swordsmen, a cut): a typed cell is what the next lab fight uses.

Who survives · the lab's levels 45 · 47 · 48 · 50

How many of that blow in a row KO a fresh man at the tier's own level. **KO** = one blow. Hover a cell for the numbers.

KO one blow · 2 two · 3 three · 4 four · 5+ five or more

· a dot in the corner = measured on the game's own machine in this read · 1+1 = his shield breaks the first shard, the second kills him

	HP / BAR HITS AT THE LAB'S LEVEL				LANDINGS · CHARGES · FISH · STOMPS TO KO				RUN CONTACTS TO KO				THE CHOMP				THE SWORDSMAN'S CUT				
man	EASY 45	MEDI 47	HARD 48	MEGA 50	E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G	THE ICE
THE LAB'S STOCK ARMY																					
RIFLEMAN	22 / 7	22 / 7	22 / 7	26 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
M. GUNNER	35 / 7	35 / 7	35 / 7	40 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
BAZOOKA	40 / 7	40 / 7	40 / 7	49 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
MORTAR	20 / 7	20 / 7	20 / 7	23 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
MEDIC	18 / 6	18 / 6	18 / 6	22 / 6	7	7	4	4	4	4	4	4	4	3	2	2	3	3	3	3	KO

	HP / BAR HITS AT THE LAB'S LEVEL				LANDINGS · CHARGES · FISH · STOMPS TO KO				RUN CONTACTS TO KO				THE CHOMP				THE SWORDSMAN'S CUT				
man	EASY 45	MEDI 47	HARD 48	MEGA 50	E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G	THE ICE
FLAME	51 / 7	51 / 7	51 / 7	61 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
SWEEPER	20 / 7	20 / 7	20 / 7	24 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
RADIO MAN	20 / 7	20 / 7	20 / 7	24 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
SARGE · SABER	32 / 12	32 / 12	32 / 12	35 / 12	13	13	7	7	7	7	7	7	7	5	4	4	5	5	5	5	KO
A TAP AWAY (THE PLATOON, SARGE'S OTHER IRONS)																					
GRENADIER	37 / 7	37 / 7	37 / 7	45 / 7	8	8	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
COMMANDO	49 / 8	49 / 8	49 / 8	58 / 8	9	9	5	5	5	5	5	5	KO	KO	KO	KO	KO	KO	KO	KO	1+1
SPY	22 / 6	22 / 6	22 / 6	27 / 6	7	7	4	4	4	4	4	4	KO	KO	KO	KO	KO	KO	KO	KO	KO
SARGE · PISTOL	29 / 8	29 / 8	29 / 8	35 / 11	9	9	5	6	5	5	5	6	5	3	3	3	3	3	3	4	KO
SARGE · SHOTGUN	32 / 8	32 / 8	32 / 8	35 / 8	9	9	5	5	5	5	5	5	5	3	3	3	3	3	3	3	KO
SARGE · SNIPER	23 / 8	23 / 8	23 / 8	26 / 8	9	9	5	5	5	5	5	5	5	3	3	3	3	3	3	3	KO
SARGE · CLAYMORE	67 / 23	68 / 23	68 / 23	68 / 23	25	25	13	13	13	13	13	13	13	9	7	7	9	9	9	9	KO

LANDINGS · CHARGES · FISH · STOMPS: the same 1 · 1 · 2 · 2 bar hits × the share, by the sum (a man mends one bar hit for every 4 s left alone, so a lone cave yeti on his 1 s dance and 1.5–3 s in the hole often needs one more). RUN: the chase's 2 bar hits every 0.5 s under him, inside the mend. THE CHOMP and THE CUT: a foot soldier falls to one (the kill door); Sarge, the medic and a hero take 2 · 3 · 4 · 4 and 3 bar hits × the share. The saber Sarge's cells: his blade turns about half of every bar-hit blow (measured; the kill doors are not his to turn: he is not a foot soldier). THE ICE: one shard, anyone (the commando's shield breaks the first over him).

The heroes at the lab's level (one in command), measured

hero	HP (E · M · H · G)	bar hits	a landing	a cut	THE CHOMP	the run	the stomp	THE ICE
KING YETI	**115 · 115 · 116 · 122**	108 · 108 · 108 · 114	every cave creature's blow bounces off his hide: the landing, the charge, the fish, the cut, THE CHOMP, the run, the stomp land nothing (measured on every tier, both engines: 56 of 56 blows turned)					KO
KING REX	**167 · 168 · 169 · 176**	99 · 99 · 99 · 105	every cave creature's blow bounces off his hide: the landing, the charge, the fish, the cut, THE CHOMP, the run, the stomp land nothing (measured on every tier, both engines: 56 of 56 blows turned)					KO
KING CROAK	**193 · 194 · 194 · 201**	159 · 159 · 159 · 165	1 · 1 · 2 · 2 bar hits	3 · 3 · 3 · 3 bar hits	2 · 3 · 4 · 4 bar hits	2 a contact	1 · 1 · 2 · 2 bar hits	KO
JANE	**101 · 101 · 102 · 108**	50 · 50 · 50 · 56	1 · 1 · 2 · 2 bar hits	3 · 3 · 3 · 3 bar hits	2 · 3 · 4 · 4 bar hits	2 a contact	1 · 1 · 2 · 2 bar hits	KO
EMPEROR BEAR (the hero)	**175 · 176 · 177 · 184**	99 · 99 · 99 · 105	1 · 1 · 2 · 2 bar hits	3 · 3 · 3 · 3 bar hits	2 · 3 · 4 · 4 bar hits	2 a contact	1 · 1 · 2 · 2 bar hits	KO
PRINCESS CROAK	**75 · 76 · 76 · 82**	59 · 59 · 59 · 65	1 · 1 · 2 · 2 bar hits	3 · 3 · 3 · 3 bar hits	2 · 3 · 4 · 4 bar hits	2 a contact	1 · 1 · 2 · 2 bar hits	KO

A hero's share is 1 (his HP is not on the unit sheet), so a hero takes the whole bar hit: a landing 1 · 1 · 2 · 2, a cut 3, THE CHOMP 2 · 3 · 4 · 4, the run 2, the stomp 1 · 1 · 2 · 2. **King Yeti and King Rex take none of them**: every cave creature is thrown snow to their hide, and it bounces (measured on every tier, both engines). The falling ice kills every hero in one (measured, all six, every tier).

The level a man first survives

For the bar-hit blows. The three kill doors have no such level: no rank survives a cut or THE CHOMP as a foot soldier, and nobody survives the ice. **Green** = he already does at the lab's EASY level 45; — = not even at rank 100.

			TWO-BAR-HIT BLOWS IN A ROW (MEGA's landing · charge · fish · stomp; the run)				MEGA's CHOMP (Sarge, the medic)		THE CUT (Sarge, the medic)	
man	HP / bar at 45	HP / bar at 100	2	4	6	8	1	2	1	2
RIFLEMAN	22 / 7	45 / 8	**5**	100	—	—	—	—	—	—
M. GUNNER	35 / 7	65 / 8	**5**	100	—	—	—	—	—	—
BAZOOKA	40 / 7	92 / 8	**6**	100	—	—	—	—	—	—
MORTAR	20 / 7	40 / 8	**6**	100	—	—	—	—	—	—
MEDIC	18 / 6	40 / 7	**10**	—	—	—	**10**	—	**8**	**42**
FLAME	51 / 7	110 / 8	**5**	100	—	—	—	—	—	—

man	HP / bar at 45	HP / bar at 100	TWO-BAR-HIT BLOWS IN A ROW (MEGA's landing · charge · fish · stomp; the run)				MEGA's CHOMP (Sarge, the medic)		THE CUT (Sarge, the medic)	
			2	4	6	8	1	2	1	2
SWEEPER	20 / 7	42 / 8	**6**	100	—	—	—	—	—	—
RADIO MAN	20 / 7	42 / 8	**6**	100	—	—	—	—	—	—
GRENADIER	37 / 7	84 / 8	**6**	100	—	—	—	—	—	—
COMMANDO	49 / 8	102 / 9	**0**	**42**	—	—	—	—	—	—
SPY	22 / 6	50 / 7	**10**	—	—	—	—	—	—	—
SARGE · SABER	32 / 12	50 / 13	**0**	**0**	**42**	—	**0**	**0**	**0**	**0**
SARGE · PISTOL	29 / 8	50 / 12	**0**	**42**	100	—	**0**	**42**	**0**	**10**
SARGE · SHOTGUN	32 / 8	50 / 9	**0**	**42**	—	—	**0**	**42**	**0**	**10**
SARGE · SNIPER	23 / 8	40 / 9	**0**	**42**	—	—	**0**	**42**	**0**	**10**
SARGE · CLAYMORE	67 / 23	78 / 24	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Read from the game's own HP and bar ladders at every rank 0–100 (the tenth-rank extras included; no gifts, soul or gear), by the sum of the blows with no mend between. "Two-bar-hit blows" = MEGA's landing, charge, fish or stomp and the run on any tier: 2 × the share each. Sarge's rows read the iron at the same rank.

The units

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags (E · M · H · G)
 **THE CAVE YETI** 6 HP × the tier · 2 · 2 · 3 · 3 an opening burst, five bursts from 8 · 6 · 5 · 4 s by tier (on EASY 8 · 13.5 · 19 · 24.5 · 29 s), the three holes nearest the toy box asleep till 25 · 20 · 15 · 10 s; THE BACK HOLE one trip in four; the roar wakes 2 more; at most 10 live	**THE LANDING** out of a hole, a 0.4 s crouch at the lip (the tell), then one bound at 430 px/s onto the nearest man within 460 px; a man downed = THE DANCE over him (1 s, a double blow while he dances), then back into the nearest hole	**1 · 1 · 2 · 2 bar hits** × 90 % + 10 % × bar ÷ HP on every man inside 34 px of where he lands, and the bowl-over (50 px) · the saber Sarge turns half (measured 4–7 of 10)	6	9	13	16	10	10	15	15	0	40 · 36 · 32 · 30

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags (E · M · H · G)
 **THE SPEARMAN** 10 HP × the tier · the premiere at 16 · 14 · 13 · 12 s · 1 · 2 · 3 · 4 live at a time	**THE CHARGE** planted at the lip 0.45 s with forepaw dust (the tell), his heading locked on the nearest man within 460 px, then straight down the line at 380 px/s and 100 px past him; a rim or the edge stops him (1.6 s planted: the shot); a man downed = the dance	**1 · 1 · 2 · 2 bar hits** × the share on every man inside 38 px of his line, once a charge, and the bowl-over (60 px)	10	15	22	26	1	2	2	2	0	64 · 58 · 51 · 48

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags (E · M · H · G)
 **THE FISHER** 8 HP × the tier · the premiere at 21.5 · 19.5 · 18.5 · 17.5 s · 1 · 1 · 2 · 2 live at a time	**THE FISH** planted at the lip, a fish every 2.6 s at the nearest man within 420 px, lobbed where he stands now (0.9 s in the air, never homing), 3 fish then back in; a man downed = the dance	**1 · 1 · 2 · 2 bar hits** × the share on every man inside 30 px of where the fish lands, and a slip (40 px); a plane nearer than his man is his mark	8	12	18	21	1	1	2	2	0	56 · 50 · 45 · 42

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags (E · M · H · G)
 **THE DARK YETI SWORDSMAN** 18 HP × the tier · the premiere at 22 · 20 · 19 · 18 s (1 · 1 · 2 · 1) · the boss calls 1 · 1 · 2 · 1 with every second strike (ACT II: every strike) · 1 · 2 · 3 · 2 live at a time	**THE LUNGE · THREE CUTS** a crouch at the lip (0.35 s), one bound to 170 px short of a man within 560 px, a 0.22 s wind-up (the steel ring), a flight at 760 px/s up to 280 px, then **three cuts 0.16 s apart**, each the nearest man not yet cut within 62 px in front; 2 lunges a trip; a man downed = the dance	**KO** to a foot soldier (rifleman, M. gunner, bazooka, mortar, grenadier, sweeper, flame, radio man, commando, spy): the kill door, whatever his HP · **3 bar hits** × the share to Sarge, the medic, the machines and a hero · the saber Sarge turns half (measured 2–8 of 10) · a pet's shield, a commando's roof or a blue wall turns the kill into 1 · 1 · 2 · 2 bar hits	18	27	40	48	1	1	2	1	0	120 · 108 · 96 · 90

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags (E · M · H · G)
 **THE FEROCIOUS YETI** his bar 450 × 2.4 · 2.8 · 1.9 · 2.6 × the raid curve · leaks 8 lives if he crosses the toy box · arrives at 25 s	**see move by move below** the hunt out of the holes (THE CHOMP, THE RAMPAGE, THE STOMP), THE CAVE-IN under 60 % of his bar, THE CHASE under 25 %	**KO** to a foot soldier under his landing, **KO** to everyone under his falling ice; bar hits to the rest	1080	1890	1881	3089	1	1	1	1	8	300 · 300 · 300 · 300

HP = each body as the caves' own spout rows bear it on that tier (measured in this read: the raid curve ×1 · 1.5 · 2.2 · 2.64 on 6 · 10 · 8 · 18; the boss's bar = the lab's sheet). n = bodies named in the solo sheet (one commander): the five opening bursts of cave yetis and the premieres of the spearmen, fishers and swordsmen; the boss's roar calls more all fight (2 cave yetis a wake, the swordsmen with his strikes, the raiders to the box). leak = lives it costs at the toy box (a cave yeti's bite 1). tags = dog tags paid when it falls on each tier, measured as born: the cave bounty × 4 · 3.6 · 3.2 · 3 since Nov 1 (the caves' purse); the boss's 300 untouched.

The Ferocious Yeti, move by move

move	EASY	MEDIUM	HARD	MEGA	how it works
THE ARRIVAL	25 s	25 s	25 s	25 s	He shows himself in the far mouth 25 s into the fight (two eyes in the black, a roar), waits 1.6 s, then vanishes into the holes and the hunt begins. Before him the holes have already fielded the cave yetis (five bursts from 8 · 6 · 5 · 4 s by tier), the spearmen (16 · 14 · 13 · 12 s), the fishers (21.5 · 19.5 · 18.5 · 17.5 s) and the swordsmen (22 · 20 · 19 · 18 s); the three holes nearest the toy box stay shut till 25 · 20 · 15 · 10 s.

move	EASY	MEDIUM	HARD	MEGA	how it works
THE HUNT	1 × the dark · the eyes 0.6 s	0.9 × the dark · the eyes 0.45 s	0.8 × the dark · the eyes 0.35 s	0.7 × the dark · the eyes 0.25 s	ACT I: between strikes he is in the dark (the tier's share of the dark; HARD and MEGA also hold every man within 300 px of his roar for 1 s). Two ice-blue eyes light in the hole he will come out of for the tier's warn, then he bounds out of the hole nearest the man deepest in your line (every third strike out of THE BACK HOLE behind you), a 0.55 s flight that tracks the man's feet, and lands. The landing is one of three on a seven-strike rota: THE RAMPAGE 4 of 7, THE STOMP 2, a single CHOMP 1.
THE CHOMP	**KO** · 2 bar hits	**KO** · 3 bar hits	**KO** · 4 bar hits	**KO** · 4 bar hits	His landing, 58 px wide: **every foot soldier under it is KO'd** whatever his HP (the kill door; measured on every tier: the rifleman, M. gunner, bazooka, flame, grenadier and commando fell). Sarge, the medic, the jeep, the tank and a hero take 2 · 3 · 4 · 4 bar hits × the share and the bowl-over (80 px) — measured: the saber Sarge turned 2–5 of 10, the pistol and the Claymore Sarge took the bar hits, King Yeti and King Rex turned every one (their hide). A man under him when he lands is the whole law; the chomp itself is the bite over the man he downed (THE DANCE).
THE RAMPAGE	3 bounds	4 bounds	5 bounds	6 bounds	Four strikes of seven: the landing above, then 2 · 3 · 4 · 5 more bounds in a chain, each at the nearest man within 420 px (0.42 s a bound), each landing **KO to the foot soldiers under it** and one bar hit lighter for the rest (1 · 2 · 3 · 3; measured: the medic took 1 on EASY under a rampage's landing, the rifleman beside him fell).
THE STOMP	1 bar hits · 150 px	1 bar hits · 150 px	2 bar hits · 150 px	2 bar hits · 150 px	Two strikes of seven: he lands, rises 0.35 s (the amber ring: the tell), and the shockwave takes 1 · 1 · 2 · 2 bar hits × the share off every man within 150 px and bowls him 130 px (measured on every man at 60 px; a shorter jig after).
THE FROST BREATH	off	off	off	off	In the file, off the rota: the cone (330 px, 1 bar hit and a freeze of 0.7 · 0.9 · 1.3 · 1.6 s) is never dealt in a fight today (Wyatt 9/26: the animation was never finished). Not measured; nothing to balance.
THE FALLING ICE	every 7 s · 2 clusters	every 6 s · 3 clusters	every 12 s · 2 clusters	every 12 s · 2 clusters	From his arrival, every 7 · 6 · 12 · 12 s a volley of 2 · 3 · 2 · 2 clusters of three shadows, each cluster aimed at a living man on his feet (its first shadow on him). A shadow grows 1.8 s (the premiere's 2.2 s), then the shard lands and **kills every unit within 40 px**: Sarge of every iron, the jeep and the tank included (measured on every tier, both engines: the rifleman fell 1 s after the shadow). A pet's shield, Francis's block or the blue wall turns a shard into 3 bar hits; the champion's grace once a life. **A commando's shield breaks the first shard** over him or over any man within 120 px of him (his one charge; measured: the rifleman in his circle stood, the commando stood, the shield spent; a second shard over the commando killed him). The ice falls only while the lights are up: inside THE CAVE-IN it is the band's.
THE CAVE-IN	9 s · 50 % dark	13 s · 40 % dark	16 s · 30 % dark	18 s · 25 % dark	ACT II, under 60 % of his bar: he runs for the mouth, the lights go down, and a BAND of falling ice walks the floor from x 1780 to 40 px outside the toy box's ring in 9 · 13 · 16 · 18 s: rows 64 px apart, 8 seats a row down the floor, a share of the seats left dark (50 % · 40 % · 30 % · 25 %: the gaps a man can stand in), each shadow the storm's 1.8 s tell and the same kill (measured: a band shard KO'd a rifleman 1.817 s after its shadow). The lights come up at GATE TWO (THE LAST MARCH: his two swordsmen burst out beside him) and the hunt resumes.

move	EASY	MEDIUM	HARD	MEGA	how it works
THE RAVAGE	—	—	under 26 %	under 26 %	HARD and MEGA only, between the cave-in and the chase: his charges down the hole columns (from the top holes down) and the sweep from the mouth to the back hole, with the swarm (2 · 3 · 5 · 7 cave yetis on the rail); his contact on the run is THE CHASE's (2 bar hits). Not staged on this page: the read's cells.
THE CHASE	210 px/s	165 px/s	190 px/s	210 px/s	ACT III, under 25 % of his bar: out of the mouth and on the floor for good, running at a man at 140 · 165 · 190 · 210 px/s; every man within 40 px takes **2 bar hits × the share** and a bowl-over every 0.5 s he stays under him (measured on every man and hero; King Yeti and King Rex turn it). Every 6 s a roar calls the raiders to the toy box and the swordsmen out. The run is the only time his bar is on the floor to shoot the whole way.
THE GATES	60 % · 25 %	60 % · 25 %	60 % · 26 % · 25 %	60 % · 26 % · 25 %	On his bar: THE CAVE-IN under 60 %; on HARD and MEGA THE RAVAGE under 26 %; THE CHASE under 25 %. Act II's strikes call the cave yetis (2 a wake) and the swordsmen with every strike (Act I: every second).
HIS BAR (one commander, the lab's level)	**1,080**	**1,890**	**1,881**	**3,089**	450 × 2.4 · 2.8 · 1.9 · 2.6 (his own multiplier by tier, re-priced Oct 8 for a fight with no dark and cut 20 % Oct 3) × the raid curve ×1 · 1.5 · 2.2 · 2.64. No rung price in the caves. A party grows it ×(1 + 0.9 a commander past the first): the table below.
THE HOLES	7	7	7	7	Three in the top wall, three in the bottom, and THE BACK HOLE behind your line (x 236, y 214: 130 is the toy box's x). A cave yeti takes the back hole one trip in four; the boss every third strike. The far mouth (x 1840) is his door alone.
THE TOY BOX	–8 lives if he crosses it	–8 lives if he crosses it	–8 lives if he crosses it	–8 lives if he crosses it	He never marches the box in the hunt: his legs run at your men. The cave yetis do (THE RAIDERS after the first minute, the roar's callers in THE CHASE): a bite costs 1 life.

His bar by the number of commanders

	1 commander	2 commanders	3 commanders	4 commanders	5 commanders	10 commanders
EASY	1,080	2,052[m]	3,024[m]	3,996	4,968[m]	9,828
MEDIUM	1,890	3,590[m]	5,292[m]	6,994	8,694[m]	17,198
HARD	1,881	3,574[m]	5,267[m]	6,960	8,653[m]	17,117
MEGA	3,089	5,868[m]	8,648[m]	11,430	14,209[m]	28,109

The game's own sheet for one to ten commanders at the lab's level: 450 × the tier × the raid curve × (1 + 0.9 a commander past the first); [m] = the boss born in a party of that size was measured at exactly this bar on every tier, both engines. No rung price rides the caves (the Pillow Fields' law is not here).

The falling ice, measured (one shard over each unit, every tier)

	who fell
EASY	7 of 7 fell (rifleman, m. gunner, bazooka, flame, medic, grenadier, the saber Sarge) · the pistol Sarge fell · the Claymore Sarge fell · the jeep wrecked · the tank wrecked · the commando: the first shard broke on his shield, the second killed him · the roof: the rifleman in his circle stood, the shield spent
MEDIUM	7 of 7 fell (rifleman, m. gunner, bazooka, flame, medic, grenadier, the saber Sarge) · the pistol Sarge fell · the Claymore Sarge fell · the jeep wrecked · the tank wrecked · the commando: the first shard broke on his shield, the second killed him · the roof: the rifleman in his circle stood, the shield spent
HARD	7 of 7 fell (rifleman, m. gunner, bazooka, flame, medic, grenadier, the saber Sarge) · the pistol Sarge fell · the Claymore Sarge fell · the jeep wrecked · the tank wrecked · the commando: the first shard broke on his shield, the second killed him · the roof: the rifleman in his circle stood, the shield spent
MEGA	7 of 7 fell (rifleman, m. gunner, bazooka, flame, medic, grenadier, the saber Sarge) · the pistol Sarge fell · the Claymore Sarge fell · the jeep wrecked · the tank wrecked · the commando: the first shard broke on his shield, the second killed him · the roof: the rifleman in his circle stood, the shield spent

What changes from EASY to MEGA

	EASY	MEDIUM	HARD	MEGA
A landing, a charge, a fish, the stomp (bar hits × the share)	1	1	2	2
THE CHOMP on Sarge, the medic, the machines, a hero (bar hits)	2	3	4	4
THE CHOMP and a cut on a foot soldier · THE FALLING ICE on anyone	KO			
The run (bar hits) · a cut on Sarge, the medic, a hero (bar hits)	2 · 3			
Body HP (the cave creatures: the raid curve)	×1	×1.5	×2.2	×2.64
His bar at the lab's level (one commander)	1,080	1,890	1,881	3,089
His bar with 2 commanders	2,052	3,590	3,574	5,868
Cave yetis an opening burst (five bursts)	2	2	3	3
Spearmen · fishers · swordsmen live at a time	1 · 1 · 1	2 · 1 · 2	3 · 2 · 3	4 · 2 · 2
Bodies in the sheet (the premieres and the bursts)	13	14	21	20
His strike clock · his share of the dark · the eyes' warn	2.4 s · ×1 · 0.6 s	2 s · ×0.9 · 0.45 s	1.7 s · ×0.8 · 0.35 s	1.5 s · ×0.7 · 0.25 s
The roar's hold (every man within 300 px)	0 s	0 s	1 s	1 s
The falling ice: every · clusters a volley	7 s · 2	6 s · 3	12 s · 2	12 s · 2

	EASY	MEDIUM	HARD	MEGA
THE RAMPAGE's bounds	3	4	5	6
THE CAVE-IN: the band's walk · the seats left dark	9 s · 50 %	13 s · 40 %	16 s · 30 %	18 s · 25 %
THE RAVAGE	—	—	on	on
THE CHASE's pace	140 px/s	165 px/s	190 px/s	210 px/s
The first burst · the three near holes wake	8 s · 25 s	6 s · 20 s	5 s · 15 s	4 s · 10 s
Dog tags a cave creature pays (× the cave bounty)	×4	×3.6	×3.2	×3
THE CAVE-IN's refund on a man it kills	×1	×1	×0.75	×0.5
The lab's stock army	level 45	level 47	level 48	level 50
MEGA's door	—	—	—	the fight's own ladder climbed

The wave sheet (solo, one commander)

EASY · 13 bodies on the sheet + the calls

1. **8 s** · 2 cave yetis out of the holes (720, 62), (720, 660)
2. **13.5 s** · 2 cave yetis out of the holes (1080, 660), (720, 62)
3. **19 s** · 2 cave yetis out of the holes (1080, 62), (720, 62)
4. **24.5 s** · 2 cave yetis out of the holes (720, 62), (1080, 62)
5. **29 s** · 2 cave yetis out of the holes (720, 62), (1080, 660)
6. **16 s** · 1 spearman out of the holes (720, 62)
7. **21.5 s** · 1 fisher out of the holes (720, 62)
8. **22 s** · 1 swordsman out of the holes (720, 62)
9. **25 s** · 1 THE FEROCIOUS YETI in the far mouth (bar ×2.4)

MEDIUM · 14 bodies on the sheet + the calls

1. **6 s** · 2 cave yetis out of the holes (720, 62), (720, 660)
2. **11.5 s** · 2 cave yetis out of the holes (1080, 660), (720, 62)
3. **17 s** · 2 cave yetis out of the holes (1080, 62), (720, 62)
4. **22.5 s** · 2 cave yetis out of the holes (360, 660), (360, 62)
5. **27 s** · 2 cave yetis out of the holes (720, 62), (1080, 660)
6. **14 s** · 2 spearmen out of the holes (720, 62), (720, 660)
7. **19.5 s** · 1 fisher out of the holes (720, 62)
8. **20 s** · 1 swordsman out of the holes (720, 62)

HARD · 21 bodies on the sheet + the calls

1. **5 s** · 3 cave yetis out of the holes (720, 62), (720, 660), (1080, 62)
2. **10.5 s** · 3 cave yetis out of the holes (1080, 660), (720, 62), (1080, 62)
3. **16 s** · 3 cave yetis out of the holes (1080, 62), (236, 214), (360, 62)
4. **21.5 s** · 3 cave yetis out of the holes (360, 660), (360, 62), (720, 660)
5. **26 s** · 3 cave yetis out of the holes (720, 62), (1080, 660), (236, 214)
6. **13 s** · 2 spearmen out of the holes (720, 62), (720, 660)
7. **18.5 s** · 2 fishers out of the holes (360, 62), (1080, 62)
8. **19 s** · 2 swordsmen out of the holes (720, 62), (720, 660)

9. **25 s** · 1 THE FEROCIOUS YETI in the far mouth (bar ×2.8)

9. **25 s** · 1 THE FEROCIOUS YETI in the far mouth (bar ×1.9)

MEGA · 20 bodies on the sheet + the calls

1. **4 s** · 3 cave yetis out of the holes (720, 62), (720, 660), (1080, 62)
2. **9.5 s** · 3 cave yetis out of the holes (1080, 660), (720, 62), (1080, 62)
3. **15 s** · 3 cave yetis out of the holes (1080, 62), (236, 214), (360, 62)
4. **20.5 s** · 3 cave yetis out of the holes (360, 660), (360, 62), (720, 660)
5. **25 s** · 3 cave yetis out of the holes (720, 62), (1080, 660), (236, 214)
6. **12 s** · 2 spearmen out of the holes (720, 62), (720, 660)
7. **17.5 s** · 2 fishers out of the holes (360, 62), (1080, 62)
8. **18 s** · 1 swordsman out of the holes (720, 62)
9. **25 s** · 1 THE FEROCIOUS YETI in the far mouth (bar ×2.6)

One wave on the clock: the holes field the cave yetis in five bursts, the spearmen, the fishers and the swordsmen at their premieres, and THE FEROCIOUS YETI shows in the far mouth at 25 s. On top of it all fight: his roar's wakes (2 cave yetis), his strikes' calls (the swordsmen), the cave swarm every 50 · 60 · 60 · 60 s, the raiders' rush every 40 · 30 · 30 · 30 s after the first minute, and the live caps (1–4 spearmen, 1–2 fishers, 1–3 swordsmen, 10 cave yetis; a party adds one).

How this was read and checked

The live client v29.00 (index.html md5 3acb3f01b6903832226913accaff9ab7, HEAD's own copy) was opened headless on this Mac and read with the game's own functions: every table, every tier's sheet at the lab's own level, every man's HP and bar at every rank, his bar for 1 to 10 commanders, the lab's 28 cells for the caves. Then each blow was **measured on the real machines** in a BOSS LAB fight on every tier with the stock army worn (level 45 · 47 · 48 · 50), every cave creature born through the game's own door and the boss through his own spout row: a cave yeti's landing on eight men (and ten on the saber Sarge for his block), a spearman's charge on eight, a fisher's fish on eight, a swordsman's three cuts on a flame, a rifleman and a medic together and one cut on each of seven men, the saber Sarge (ten tries), the pistol and the Claymore Sarge, THE CHOMP on eight men, the pistol and the Claymore Sarge, the saber Sarge ten times, the jeep and

the tank, and a rampage's lighter landing, the chase's run on eight, the stomp on eight, a falling shard over eight men, the pistol and the Claymore Sarge, the jeep and the tank, a band's shard, a commando twice (his shield, then his death) and the roof over a rifleman, the six heroes under all eight blows, every body as the tier bears it, and the boss born in a party of 2, 3 and 5. Chrome 381/381 · WebKit 381/381 checks passed, zero page errors, and the two engines read every number the same. The opening's own clock (the first cave yeti, the box bitten with nothing placed, the lab remembering ×4) is CARD-ICN's read of v28.96, on its page; since then the quiet start moved the first burst to 8 · 6 · 5 · 4 s by tier.

Sources (the code's own names): RAID_INFO.caves, cavesCfg's spout rows (cyeti · lyeti · hyeti · dyeti · fyeti; hpX CAVES_HPX), RAID_HP_MUL, BLB_STOCK_LV, blbStockDef; FDEF rows (sno:1 on the cave family → HERO_REFL's bounce in tanHit); cyLand (CY_PIPS, CY_HIT_R, CY_SHOVE) · updateFoeSyeti's charge (SY_PIPS, SY_LANE) · hyLand (HY_PIPS, HY_R) · dyCut / dyCutHit (DY_CUT_R, DY_CUT_N, DY_CUT_C, DY_FOOT → knockKill; DY_CUT_HEAVY, DY_CUT_PIPS) · fyLand (FY_LAND_R, FY_PIPS, FY_RAMP_PIPS_DN, FY_CHOMP_SHOVE; DY_FOOT → knockKill) · updateFoeFyeti's run (FY_RUN_R, FY_RUN_PIPS, FY_RUN_HIT_CD, FY_RUN) · fyStomp (FY_ST_R, FY_ST_PIPS, FY_ST_SHOVE) · fyBreathTick (off FY_ROTA) · fyStorm / hrtBandRow → caveIcPass → caveDropHit (FY_ICICLE_R, FY_STORM_WARN, FY_ICICLE_PIPS → knockKill) · cirCatch (ENH_CMDO_GUARD_R, cmdoCharged) · fyAct (FY_ACT2, FY_ACT_RV, FY_ACT3), FY_ROTA, FY_RAMP_N, HRT_CAVE_TT, HRT_BAND_*, CAVES_* cells; tanHit (DSM k 0.9, dsmX / dsmH, gritCap, the 4 s mend); blbSheet('caves', tier, N), bhpX; bfHpAt, usBfHP, sgBfHP, heroBfHP, saberBlocks. Kit: design/boss-caves-unit-breakdown-1101/ (kit/readCaves.mjs · kit/proveCaves.mjs · kit/build.py · kit/render.mjs · kit/export_dash.py; data/; proof/logs/). Zero game bytes.